Speed Revision

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Disclaimer: This paper can contain irrelevant material.

1 Logic & Proof Techniques

1.1 Implication & Equivalences

- **Standard Implication:** $P \to Q \equiv \neg P \vee Q$.
- **Contrapositive:** $P \to Q \equiv \neg Q \to \neg P$.
  - *Application:* Used to prove "Convergent $\implies$ Bounded" by showing "Not Bounded $\implies$ Not Convergent".
- **Negation:** $\neg(P \to Q) \equiv P \wedge \neg Q$.
- **Quantifiers:**
  - Negation of $\forall x \exists y, P(x, y)$ is $\exists x \forall y, \neg P(x, y)$.
  - "Smallest positive real" logical form: $\forall x \exists y (y < x)$ implies no smallest, versus $\exists x \forall y (x \leq y)$.

1.2 Induction Skeletons

- **Standard Induction:**
  1. **Base Case:** Verify $P(1)$.
  2. **Hypothesis:** Assume $P(k)$ is true.
  3. **Inductive Step:** Show $P(k) \implies P(k+1)$ (or $P(k+2)$ depending on step size).
  4. **Conclusion:** True for all n by PMI.
- **Inequality Induction:**
  - *Example:* Proving $2^n < n!$ for $n \geq 4$.
  - *Technique:* Manipulate one side to match the other: $2^{k+1} = 2 \cdot 2^k < 2 \cdot k! < (k+1)k!$.
- **Strong Induction:** Uses $\left(\bigwedge_{i=1}^{k} P(i)\right) \implies P(k+1)$.

2 Set Theory & Functions

2.1 Cardinality

- **Countable:** Finite sets, $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$, finite subsets of $\mathbb{N}$ (e.g., finite bit strings), $\mathbb{Z} \times \mathbb{Z}$.

- **Uncountable:** $\mathbb{R}, (0,1)$, infinite bit strings, functions $f : \mathbb{N} \to \{0,1\}$.
  - *Cantor-Schröder-Bernstein (CSB):* If there exist injections $f : A \to B$ and $g : B \to A$, then $|A| = |B|$. Used to prove $|(0,1)| = |[0,1]|$.
- **Power Set:** If $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.

2.2 Function Properties

- **Injectivity (One-to-One):**
  - *Definition:* $f(x) = f(y) \implies x = y$.
  - *Calculus Check:* If $f'(x) > 0$ (strictly increasing), f is injective.
- **Surjectivity (Onto):**
  - *Definition:* For every $y \in$ Codomain, $\exists x$ such that $f(x) = y$.
- **Composition:** If $g \circ f$ is injective, f (inner) must be injective.
- **Inverse Images:**
  - Identity: $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$.
  - *Note:* f is injective $\iff f(A \cap B) = f(A) \cap f(B)$ for all subsets.

3 Relations

3.1 Properties

- **Reflexive:** $\forall x, (x,x) \in R$.
- **Symmetric:** $(x,y) \in R \implies (y,x) \in R$.
- **Transitive:** $(x,y) \in R \wedge (y,z) \in R \implies (x,z) \in R$.
- **Antisymmetric:** $(x,y) \in R \wedge (y,x) \in R \implies x = y$.

3.2 Equivalence Relations

- Must be Reflexive, Symmetric, and Transitive.
- *Example:* $xRy \iff 4 \mid (3x + y)$ is an equivalence relation.

4 Number Theory

4.1 Modular Arithmetic

- **Properties:**
  - If $a \equiv b \pmod{m}$, then $a^k \equiv b^k$ and $ac \equiv bc$.
  - **Zero Divisors:** Elements $a \neq 0$ where $ab \equiv 0 \pmod{n}$ (e.g., 4 in $\mathbb{Z}_{12}$).
  - **Units:** Elements coprime to n have inverses (e.g., $\{1, 3, 7, 9\}$ in $\mathbb{Z}_{10}$).

4.2 Solving Equations

- **Linear Congruences:** $ax \equiv b \pmod{n}$. Solvable if $\gcd(a, n) \mid b$.
- **Chinese Remainder Theorem (CRT):**
  - System: $x \equiv a \pmod{n}, \quad x \equiv b \pmod{m}$ (where $\gcd(n, m) = 1$).
  - *Method:* Express $x = nk + a$, substitute into second equation.
- **Diophantine Impossibility:**
  - To prove no integer solutions for $15x^2 - 7y^2 = 9$:
  - Analyze modulo 3 or 5. Check quadratic residues ($x^2 \pmod{5} \in \{0, 1, 4\}$). Contradiction arises (e.g., needing $k^2 \equiv 2 \pmod{5}$).

5 Complex Numbers

5.1 Forms & Powers

- **Representations:** Rectangular $z = a + bi$, Exponential $z = re^{i\theta}$.
- **De Moivre:** $(re^{i\theta})^n = r^n e^{in\theta}$.
- **Roots of Unity:** $z^n = 1$ forms a regular polygon.

5.2 Geometry (Loci)

- **Circle:** $|z - a| = r$.
- **Perpendicular Bisector:** $|z - a| = |z - b|$.
- **Ellipse:** $|z - a| + |z - b| = k$ where $k > |a - b|$.
- **Apollonius Circle:** $|z - a| = k|z - b|$ $(k \neq 1)$.
  - *Proof:* Square both sides $|z - a|^2 = k^2|z - b|^2$ and expand using $z = x + iy$.

6 Analysis (Sequences & Limits)

6.1 Sequences

- **Boundedness:** Convergent $\implies$ Bounded.
- **Monotone Convergence:** A bounded, monotonic sequence converges.
- **Recursive Limits:**
  - Given $a_{n+1} = \sqrt{2 + a_n}$, prove bounded (< 2) and increasing.
  - Find limit L: Solve $L = \sqrt{2 + L} \implies L = 2$.

6.2 Limit Computations

- **Conjugate Trick:** For $\lim(\sqrt{n^2 + 5n} - n)$, multiply by $\frac{\sqrt{\cdots}+n}{\sqrt{\cdots}+n}$ to remove radical.
- **Factorials:** $\lim \frac{n!}{n^n} = 0$.

6.3 Formal Proofs

- **Epsilon-N $(\epsilon - N)$:**
  - Goal: $|a_n - L| < \epsilon$ for $n > N$.
  - Strategy: Bound term by $\frac{C}{n}$, set $\frac{C}{N} < \epsilon$.
- **Epsilon-Delta $(\epsilon - \delta)$:**
  - Goal: $0 < |x - c| < \delta \implies |f(x) - L| < \epsilon$.
  - *Example:* $\lim_{x \to 2} x^2 = 4$. Decompose $|x^2 - 4| = |x - 2||x + 2|$. Bound $|x + 2|$ by assuming $\delta \leq 1$.

7 Combinatorics

7.1 Counting Techniques

- **Permutations:** $n!$ for distinct items.
- **Mississippi Rule:** For indistinguishable items: $\frac{n!}{n_1!n_2!\ldots}$.
- **Stars and Bars:**
  - Integer solutions to $x_1 + \cdots + x_k = n$ is $\binom{n+k-1}{k-1}$.
  - *Constraints:* If $x_i \geq c$, substitution $y_i = x_i - c$ reduces n by $\sum c$.

7.2 Binomial Coefficients

- Coefficient of $x^k y^{n-k}$ in $(x+y)^n$ is $\binom{n}{k}$.

8 Linear Algebra & Vector Spaces

8.1 Vector Spaces

- **Subspaces:** Must contain $\mathbf{0}$, be closed under addition, and closed under scalar multiplication.
  - *Fails:* Lines not through origin ($x+y=1$), Polynomials of degree **exactly** n.

8.2 Vector Geometry

- **Dot Product:** $\mathbf{u} \cdot \mathbf{v} = 0 \iff \mathbf{u} \perp \mathbf{v}$.
- **Projection:** $\text{proj}_{\mathbf{u}}\mathbf{v} = \frac{\mathbf{v}\cdot\mathbf{u}}{|\mathbf{u}|^2}\mathbf{u}$.
- **Decomposition:** $\mathbf{v} = \mathbf{v}_{\parallel} + \mathbf{v}_{\perp}$.
- **Plane Equation:**
  - Normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.
  - Equation: $\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$.
- **Skew Lines Distance:** $D = \frac{|(\mathbf{r}_2 - \mathbf{r}_1)\cdot(\mathbf{d}_1 \times \mathbf{d}_2)|}{||\mathbf{d}_1 \times \mathbf{d}_2||}$.

9 Polynomials & Rings

9.1 Polynomial Properties

- **Vieta's Formulas:**
  - $\sum x_i = -a_{n-1}/a_n$.
  - $\sum x_i^2 = (\sum x_i)^2 - 2\sum x_i x_j$.
- **Remainder Theorem:** Remainder of $P(x) \div (x - c)$ is $P(c)$.
- **Conjugate Roots:** If coefficients are real, roots come in pairs $a \pm bi$.

9.2 Ring Theory

- **Boolean Ring:** If $a^2 = a$ for all a, the ring is commutative ($ab = ba$).

10 Computational (Julia)

- **Sets:** ‘Set([1, 2, 2])‘ results in ‘Set([1, 2])‘ (unique elements).
- **Control Flow:** ‘for x in S‘, ‘if x % 2 != 0‘ (modulo check).
- **Modifying Collections:** ‘push!(collection, item)‘ appends data.